Network Analysis of Consumer Behaviors in Qualification Courses

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Abstract

Qualifications such as CPA(Certified Public Accountant) and licensed tax accountants have a big impact on professional life. Additionally, the number of participants of these certification exams reflects temporal socioeconomic situations. However, the number of students who take qualification courses to prepare for these exams has little attention. In the present work, we take network approach to evaluate temporal popularity and robustness of qualification courses. Here we take a course as a node and students' flow as a link. We expect that our approach can apply not only to qualification courses but also for other networks.

Keyword: Network analysis, Empirical analysis, Social Modeling, Educational Service

Qualifications such as CPA(Certified Public Accountant) and licensed tax accountants have a big impact on professional life. Therefore, many people who want to pass the qualification exams go to technical college to take special courses. Furthermore, the number of people who take special courses changes due to macroscopic socioeconomic situations. For example, the number of people who prepare for civil service exams increases during economic slowdown.

However, the number of students who take courses in technical college reflect are rarely taken into account although it reflects the current society.

We use data from approximately one million students in a technical college in Japan for 18 years since 1996. Our data contain 2.9 million records including various items, such as date, course, amount, gender, occupation etc. We can also easily identify students who have taken the course before. Furthermore, our data consist of data during the Global Financial Crisis which started in 2008. Therefore, we can observe changes in students' behaviors at that time.

In this research, we apply network approach to evaluate temporal course popularity and its robustness for external factors. Here we take each course as a node and students' flow as a link. Thus, a node that has many in-link is a popular course. There exists a good method to estimate popularity by temporal changes of links with binary conditions (exist/not) [1]. However, our network can define further conditions such as the weight of the link from students' flow. Moreover, we also discuss popularity and robustness of the node by using information about students such as gender and number of repeats.

We expect that our approach will open new horizons to estimate popularity and robustness of nodes in various networks.

References

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